Workflow Summary

This project is a **Customer Churn Analysis and Customer Intelligence** study built using **Python, Pandas, SQLite, Matplotlib, and Seaborn**. The workflow started by importing three Excel sheets into SQLite as separate tables: db_customer, db_subscription, and db_support. Each sheet was then loaded back into Pandas for cleaning, transformation, analysis, and visualization. The Customerid column was used as the common key across all tables, which made it possible to perform table-level analysis and merge the data into one final analytical dataframe.

The data cleaning stage focused on standardizing the raw inputs and making them analysis-ready. In db_customer, the Dob column was converted to datetime, missing Country values were filled using the State to Country mapping, the Interests and Pincode columns were removed, and Name was renamed to Customer_name. Gender values were standardized by replacing Men with Male and Women with Female. In db_subscription, the date columns were converted to datetime format. In db_support, the Complaint_Date column was converted to datetime and the unwanted Col_1 and Comment columns were dropped.

After cleaning, the three tables were merged using Customerid. Since db_subscription was the largest table, it was used as the base table and merged with the customer and support tables. This produced a final dataframe with **650,000 rows and 21 columns**. Additional feature engineering was then applied to create Churn_flag, Customer_age, tenure_days, Churn_risk, and Cancellation_month, which helped transform the dataset from raw transactional data into a more meaningful churn analytics dataset.

Key Findings

The overall churn rate of the dataset was **24.3%**, while the retention rate was **75.7%**. This indicates that roughly one out of every four customers in the dataset was classified as churned. The average revenue per user (ARPU) was **17.51**, the average customer tenure was about **1402.10 days**, and the estimated revenue at risk from churned users was **2,223,747**.

Churn behavior differed clearly by product and plan type. The **Basic** plan had the highest churn rate at **31.61%**, followed by **Standard** at **21.88%**, and **Premium** at **12.10%**. Similarly, churn by subscription type showed the same type of pattern, with **Organic** having the highest churn rate, followed by **Paid**, and then **Refferal**. These results suggest that lower-tier or more casual customer segments were more likely to leave.

Geographic churn differences were also strong. State-level churn varied widely across regions, with some states showing very low churn and others showing much higher churn.

This indicates that churn is not evenly distributed and may be influenced by geography, customer behavior, or local market conditions. Country-level and state-level churn visualizations were used to identify these differences more clearly.

The support dataset showed that not all customers complained, which was important for making the support analysis more realistic. The support table contained **37,500 complaint records** and **30,000 unique customers**, meaning some customers raised more than one complaint. This helped support analysis remain closer to a real customer service pattern instead of assuming that every customer had an issue.

Analysis Summary

The project shows that churn can be analyzed effectively when customer, subscription, and support data are connected through a common identifier. Cleaning the raw data, fixing missing values, standardizing categories, and creating engineered features made the final analysis much more reliable and readable. The merged dataframe enabled analysis across customer demographics, subscription behavior, support interaction, and churn risk in one place.

The visual analysis was divided into two parts: **Matplotlib** and **Seaborn**. Matplotlib was used for monthly churn trends, churn by plan type, churn by country, and churn by state. Seaborn was used for correlation heatmaps, pair plots, and categorical comparisons. Finally, pivot tables were created to summarize churn by plan type, which confirmed the same pattern seen in the main analysis: **Basic customers churned the most, followed by Standard and then Premium**.

Overall, this project provides a complete churn analysis pipeline, starting from raw data ingestion and cleaning and ending with business-focused insights. It demonstrates how customer behavior, subscription structure, and support interaction can be combined to identify churn patterns and support retention strategy.